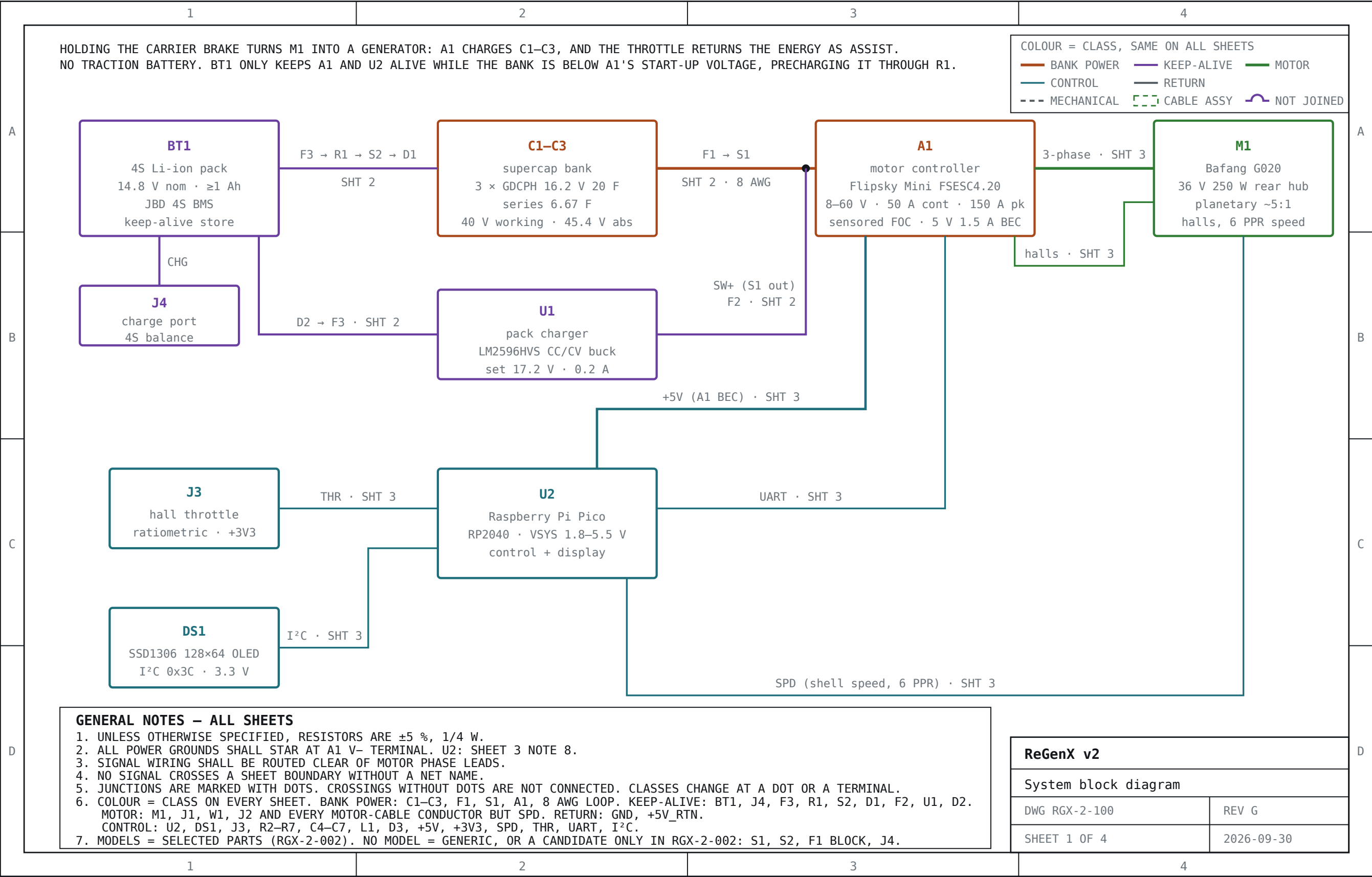
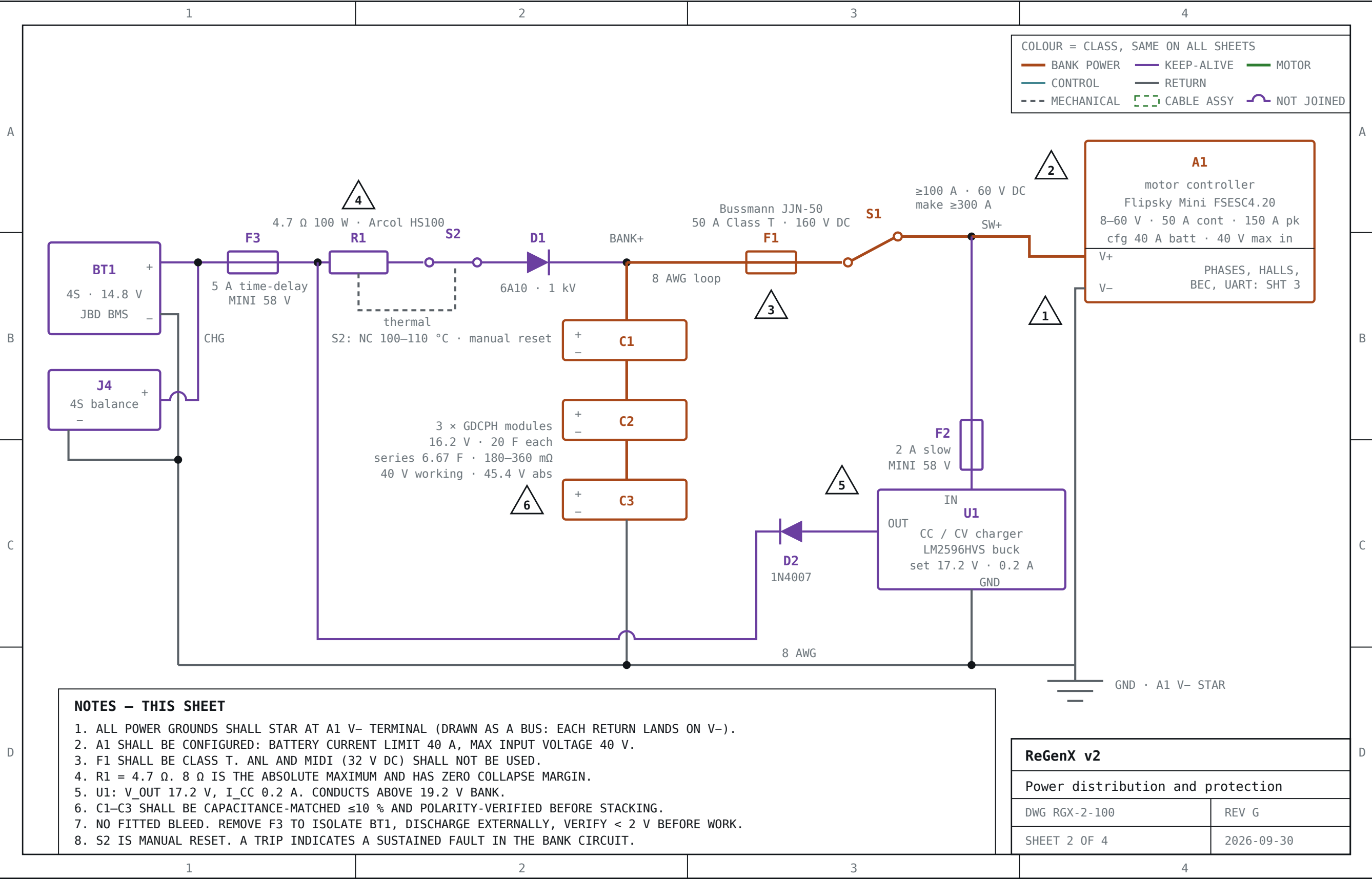


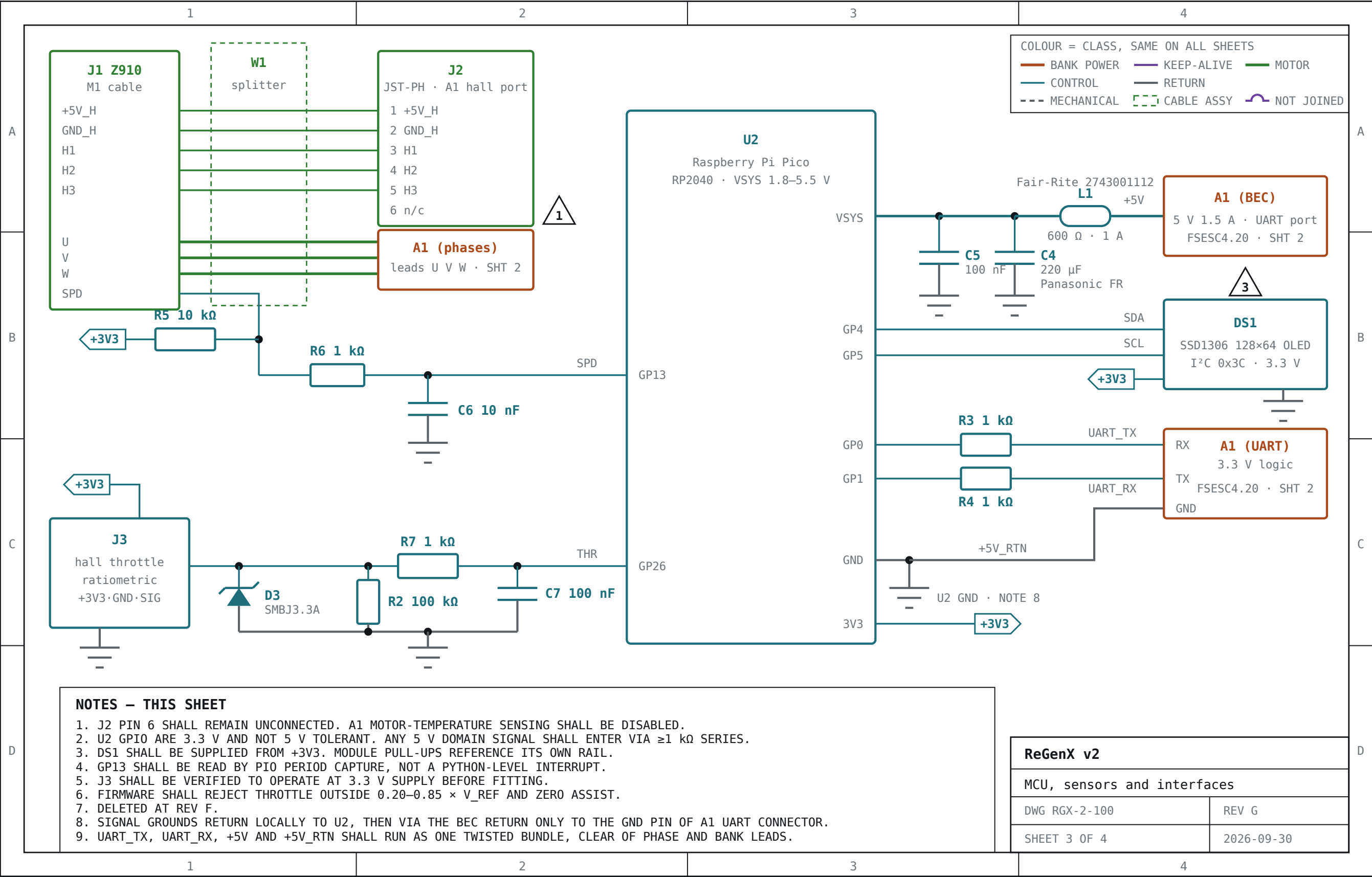
ReGenX v2 – Schematic Set

Sheet 1 of 4 – System block diagram

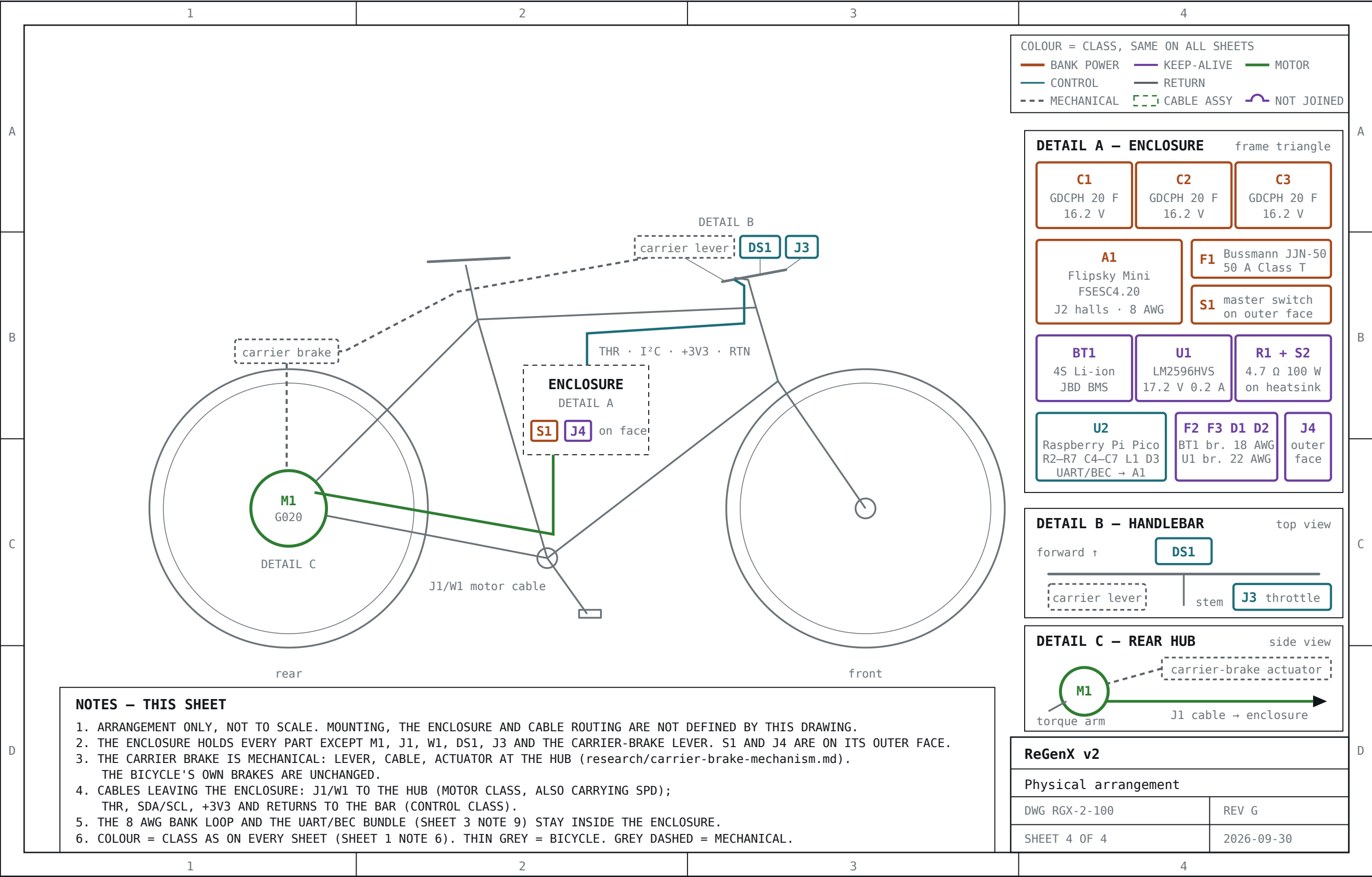




ITEM	ZONE	DESCRIPTION	SPECIFICATION
BT1	B1	Battery, secondary	4S Li-ion, 14.8 V nom, 16.8 V max, ~15 Wh. Integral BMS: balance, 0V 4.25 V/cell, UV 3.00 V/cell, 0C
J4	B1	Connector, charge	4S balance-charge input to BT1 BMS. Upstream of F3. Balance taps not drawn. P/N TBD; RGX-2-002 candidate: XH-4S balance lead
F3	B1	Fuse	5 A time-delay, ≥32 V DC, in accessible inline holder – removing F3 is the BT1 service disconnect. 18 AWG branch protection. Duty 3.36 A / 90 s
R1	B1–B2	Resistor, power	4.7 Ω ±5 %, 100 W wirewound alu-clad. Heatsink ≥150 cm² at 25 °C ambient. Duty 53 W / 22 s per cold start from a flat bank, 832 J, ΔT 31 K
S2	B2	Switch, thermal cutout	Normally-closed bimetal, open 100–110 °C, manual reset, ≥5 A, ≥50 V DC. Bonded to R1 body. P/N TBD; RGX-2-002 candidate: KSD301 (48 V DC, with the voltage line relaxed to ≥24 V DC, queued for the spec)
D1	B2	Diode, rectifier	Si, I _{F(AV)} ≥ 6 A, V _{RRM} ≥ 100 V, I _R ≤ 10 μA @ 25 °C, V _F ≈ 1.0 V @ 3.4 A. 6A10 or equiv
C1–C3	B2–C2	Capacitor, EDLC module	16.2 V, 20 F each (6 × 2.7 V 120 F). Series 6.67 F, ESR 180–360 mΩ, 40 V working, 45.4 V absolute at ≤10 % spread. Integral balancing is intra-module cell matching only – it does not divide voltage between modules
F1	B3	Fuse	50 A Class T, 160 V DC (Bussmann JJN-50 or equiv) + block. Zone of protection begins at F1. Block P/N TBD; RGX-2-002 candidate: Blue Sea 5502100
S1	A3–B3	Switch, disconnect	SPST, ≥100 A continuous, ≥60 V DC break, make ≥300 A (min ESR at 45.4 V). P/N TBD; RGX-2-002 candidate: WATERWICH 275 A marine disconnect
F2	B3–C3	Fuse	2 A slow-blow, ≥58 V DC, inline holder. 22 AWG branch protection
U1	C3	Converter, DC-DC	LM2596HVS CC/CV buck module. V _{in} 5–57 V, dropout 2 V. Set V _{out} 17.2 V, I _{CC} 0.2 A. Conducts V _{bank} ≥ 19.2 V
D2	C3	Diode, rectifier	Si, I _{F(AV)} ≥ 1 A, V _{RRM} ≥ 100 V, V _F ≈ 0.8 V @ 0.2 A. 1N4007 or equiv
A1	A4–B4	Assembly, motor controller	Flipsky Mini FSESC4.20. 8–60 V, 50 A cont / 150 A pk, sensed FOC, 5 V 1.5 A BEC, 3.3 V UART, idle 3 W. Battery current limit 40 A; maximum input voltage 40 V. 8 AWG harness terminates at the factory XT60; pigtails shall not be extended
–	B2–B4	Wire, bank circuit	8 AWG (~55 A bundled), ring terminals. Keyed or size-differentiated at the C1–C3 module ends
–	B1, C3	Wire, branch circuits	18 AWG pack branch; 22 AWG charger branch



ITEM	ZONE	DESCRIPTION	SPECIFICATION
M1	SHT 1	Motor, geared hub	Bafang G020, rear. Single-stage planetary ~5:1. Halls ×3 + shell speed 6 pulse/rev. One 9-pin Higo Z910 cable (J1), every pin used: no thermistor conductor. Thermal protection: none fitted
U2	A2–C3	Assembly, MCU board	Raspberry Pi Pico, RP2040. VSYS 1.8–5.5 V, +3V3 out ≥300 mA, load 30 mA. GPIO 3.3 V, not 5 V tolerant, integral Schmitt. ADC 12-bit ref AVDD; ADC3 = VSYS/3
J1	A1	Connector, motor cable	Higo Z910 9-way, M1 supplied: phases U, V, W; +5 V, GND, H1–H3, shell speed (white). Pin assignment by continuity map before first power (RGX-2-002 Inspection item 1): Z910 internals vary
J2	A2	Connector, controller hall	JST-PH 6-way on A1's hall port: +5V_H, GND_H, H1–H3. Pin 6 unconnected
W1	A1	Cable assembly, splitter	Higo Z910 9-pin extension, cut. Phases U, V, W to A1 phase leads; +5V_H, GND_H, H1–H3 to J2; shell-speed conductor branched to R6
R5	B1	Resistor	10 kΩ ±5 %, 1/4 W. Shell-speed pull-up to +3V3
R6	B1	Resistor	1 kΩ ±5 %, 1/4 W. Series. Fault injection limit 1.7 mA at 5 V
C6	B2	Capacitor	10 nF, ≥50 V, C0G. R6·C6 = 10 μs noise filter (a 1 μs, 5 V spike reaches the pin at ≈ 0.5 V); rising edge (R5+R6)·C6 = 110 μs; signal band ≤ 30 Hz
J3	C1	Connector, throttle	3-way. Hall throttle, ratiometric, supplied from +3V3
D3	C1	Diode, TVS	Unidirectional, V _{RWM} 3.3 V, V _C ≤ 4.5 V @ 1 A, SOD-323 or larger. U2 GPIO absolute max = IOVDD + 0.5 = 3.8 V
R2	C2	Resistor	100 kΩ ±5 %, 1/4 W. Throttle pulldown, open-circuit detection. R2·C7 = 10 ms
R7	C2	Resistor	1 kΩ ±5 %, 1/4 W. THR series. Fault injection limit 1.7 mA at 5 V. R7·C7 = 100 μs; signal band ≤ 10 Hz
C7	C2	Capacitor	100 nF, ≥16 V, X7R. ADC charge reservoir, mount at the U2 pin
R3, R4	B3, C3	Resistor	1 kΩ ±5 %, 1/4 W. UART series. Fault injection limit 1.7 mA at 5 V; rise 30 ns at 10 pF
L1	A4	Ferrite bead	Fair-Rite 2743001112, axial. ~600 Ω @ 100 MHz, I ≥ 1 A, DCR ≤ 0.1 Ω
C4	B4	Capacitor	Panasonic FR 220 μF 25 V. 220 μF ±20 % , ≥10 V, aluminium electrolytic, 105 °C, ≥2 000 h. VSYS hold-up 6.8 ms nominal / 5.5 ms at –20 %, at 0.35 W to the 1.8 V floor
C5	B3	Capacitor	100 nF, ≥16 V, X7R. Mount adjacent to U2 VSYS pin
DS1	B4	Display	SSD1306 128×64 OLED, I²C 400 kHz, 3.3 V supply, address 0x3C, ~20 mA
A1	A2, A4, C4	Assembly, motor controller	Reference SHEET 2: phase leads, hall port (J2), BEC and UART connector. BEC load 70 mA of 1500 mA
–	all	Wire, signal	24 AWG signal; the +5V / +5V_RTN pair is 22 AWG (SHEET 2 branch wire). Shell speed twisted with GND. UART_TX and UART_RX twisted with the +5V / +5V_RTN pair as one bundle, per NOTE 9



— Interface schedule

NET	FROM	TO	SHEET	TYPE
BANK+	C1 positive	F1 → S1 → A1 V+	2	power, 8 AWG
SW+	S1	A1 V+, F2	2	power, 8 AWG
GND	A1 V− star	BT1 −, C3 −, U1 −	2	power. 8 AWG C3 − leg; 18 AWG BT1 − leg; 22 AWG U1 − leg. U2 is not on this net − see +5V_RTN
+5V_RTN	U2 GND	A1 GND pin, UART connector	3	return, 22 AWG. Sole U2 ground path, per NOTE 8; bundled per NOTE 9
+5V	A1 BEC	U2 VSYS via L1, C4, C5	1, 3	power, 22 AWG; bundled per NOTE 9
+3V3	U2 3V3 out	J3, DS1, R5	3	power, 24 AWG
SPD	J1 shell speed (white)	U2 GP13 via R6	3	digital, 5 V domain at source, 6 pulse/rev, 8–24 Hz at 10–30 km/h
THR	J3 SIG	U2 GP26 (pin 31) via R7	3	analog, 0–3.3 V
THR_RTN	J3 GND	U2 GND, local	3	return, 24 AWG, per NOTE 8
DS1_RTN	DS1 GND	U2 GND, local	3	return, 24 AWG, per NOTE 8
UART_TX	U2 GP0 via R3	A1 UART RX	3	digital, 3.3 V; bundled per NOTE 9
UART_RX	A1 UART TX	U2 GP1 via R4	3	digital, 3.3 V; bundled per NOTE 9
SDA / SCL	U2 GP4 / GP5	DS1	3	I²C, 3.3 V
H1–H3	J1 H1–H3	J2 pins 3–5	3	digital, 5 V, A1 domain
+5V_H, GND_H	J2 pins 1–2 (A1 hall port)	J1	3	hall supply, 5 V, A1 domain. Not the +5V BEC net
U, V, W	J1 phases	A1 phase leads	3	power, motor cable conductors
CHG	J4 +	BT1 +, upstream of F3	1, 2	power, 18 AWG

— Revision history

REV	DATE	DESCRIPTION
A	2026-08-02	Initial release. Three sheets.
B	2026-08-02	Added J4, S2, R5, R6, C6, C7, D3. F1 → 50 A Class T. F3 → 5 A time-delay. D1 → 6 A. R3, R4 → 1 kΩ. U1 conduction threshold 19.2 V, setpoints added. Sheet 1 +5V source corrected to A1. Zone references corrected. NOTES 4–8 added.
C	2026-08-02	Added R7 in THR series. D3 respecified by V _C and redrawn as TVS. Bank circuit 10 → 8 AWG. R1 50 → 100 W, energy 650 → 832 J. S1 make 150 → 300 A, zone A3 → B3. C4 100 → 220 μF, zone A3 → B3. S2 auto → manual reset, voltage rating added. C1–C3 absolute 48.6 → 45.4 V. +5V_RTN added and U2 removed from the GND net. SHEET 2 NOTES 2, 4, 6, 8 and SHEET 3 NOTE 2 revised.
D	2026-08-02	D3 symbol orientation corrected (cathode to THR). S2 redrawn normally-closed. SHEET 2 flags 2 and 4 exchanged to match renumbered notes. Junction dots added at C3 −, U1 − and R2 tees; crossing convention stated in SHEET 1 NOTE 5. R5 branch tee moved to its junction. P/N TBD marked on S1, S2, F1 block, J4. A1 pigtail termination noted. F2 and R2 symbols rotated to match branch direction. D2 return rerouted clear of the bank stack. Polarity marks added to BT1, J4, C1–C3. Title blocks to REV D and titles matched to sheet headings. Sheet-boundary nets renamed +5V, SPD, THR. UART nets labelled; A1 pin sides marked. F3 holder designated BT1 service disconnect; NOTE 7 revised. Zone listings corrected: R1 B1–B2, F2 B3–C3, DS1 B4–C4.
E	2026-09-29	C6 1 → 10 nF: shell-speed input filtered against phase-lead PWM coupling (R6·C6 1 → 10 μs; rising edge 110 μs, about 1 % of a half-period at 60 km/h). Title blocks to REV E.
F	2026-09-29	U2 ground lands on the GND pin of A1's UART connector, the reference of the UART signals, instead of the V− star, so no battery current flows between the two grounds: +5V_RTN drawn on SHEET 3, SHEET 3 NOTE 8 revised, NOTE 9 added (UART and BEC pair as one twisted bundle), SHEET 1 NOTE 2 and SHEET 2 NOTE 1 limited to power grounds. SHEET 3 NOTE 7 (VSYS logging) deleted. J1 redrawn as M1's single Higo Z910 cable, pins by continuity map; W1 is the Z910 splitter (queued from RGX-2-002 Rev A). Sheet 3 notes respaced. Title blocks to REV F.

REV	DATE	DESCRIPTION
G	2026-09-30	Colour by class on every sheet (bank power, keep-alive, motor, control, return), legend at top right, SHEET 1 NOTES 5-7. Selected models and ratings written at each part; note flags in ink, net flags in the net's class. SHEET 1: system statement added, UART now runs to A1 (it ended at U1 in Rev F), SW+ branches from the F1/S1 link, U1 under the bank, CHG named. SHEET 2: A1 V- return routed clear of U1 (it ran behind it in Rev F), ground bus ends at the A1 V- star, J4 below BT1 so nothing crosses, S2 drawn closed with a grey thermal link, hop where the D2 return crosses the C3 ground, U1 pins named. SHEET 3: phases U V W drawn to A1, hall supply named +5V_H / GND_H, J2 → A1 hall port, A1 UART and DS1 exchanged so +5V_RTN crosses nothing, C4 and C5 clear of R3 and R4, DS1 and J3 returns drawn, +3V3 net flags. SHEET 4 physical arrangement added. Parts lists: RGX-2-002 candidates named for S1, S2, F1 block, J4; L1 and C4 models; zones. Interface schedule: +5V_H/GND_H and U V W rows, +5V_RTN on sheet 3 only. Title blocks to REV G, 4 sheets.